Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.



Team Details

a. Team name: Szen Innovation
b. Team leader name: Maria Amor Iglesia
c. Problem Statement: Empowering SMBs & Creators with Multi-Signal Opportunity Pipelines & Human-in-the-Loop Governance

Our solution

ApexGrowth AI

Autonomous Social Media Manager with Human-in-the-Loop control

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Brief about the idea

Vision

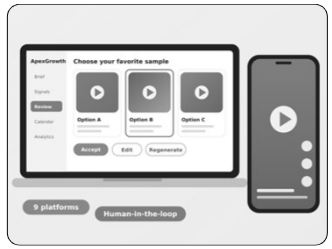
An autonomous, multimodal social media manager that helps businesses and creators discover, generate, review and distribute viral short-form content, using multi-signal intelligence and Human-in-the-Loop control.

The challenge

Small businesses and digital brands struggle to stay consistently visible, capture leads and decode platform algorithms without burning out or losing their brand authenticity.

Theme alignment

Media, Content & Digital Experiences



Discover → Generate → Review → Distribute

Opportunities

How different is it from any of the other existing ideas?

Five signals, one supervised pipeline

Trends, news, personal ideas, festivals/events and creative references are combined, so content is timely, unique and non-derivative.

How will it be able to solve the problem?

Automates the data crunching

Handles multi-source data aggregation, trend synthesis and multi-platform scheduling — saving owners 20+ hours every week.

USP of the proposed solution

Supervised autonomy

A 3-sample selection dashboard keeps brand safety, tone consistency and human editorial oversight on every post.

List of features offered by the solution

User Onboarding Brief

Creator sets niche, audience, geography, language, platforms, goals and reference styles.

Multi-Signal Discovery

Scans trends, news, events, personal ideas and references into a prioritized queue.

Multimodal Generation

Gemini turns opportunities into hooks, angles, scripts and captions, as 3 sample videos.

Human-in-the-Loop Review

Accept, Reject, Edit, Regenerate or Save each option to the content calendar.

Automated Distribution

Approved content dispatches through Blotato to 9 social platforms.

Performance Analytics

Performance metrics for every post are logged to Google Sheets.

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Process flow diagram or Use-case diagram

01 Onboarding Brief
Creator sets niche, audience, region, language, platforms and goals.

02 Opportunity Discovery
AI scans trends, news, events and ideas into a prioritized queue.

03 Multimodal Generation
Gemini creates hooks, angles, scripts and captions: 3 sample videos.

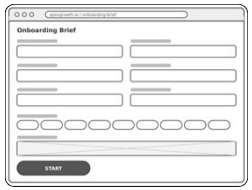
04 Human-in-the-Loop
Creator can Accept, Reject, Edit, Regenerate or Save.

05 Distribution & Analytics
Approved posts go to 9 platforms; metrics log to Google Sheets.

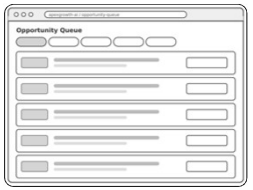
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Wireframes/Mock diagrams of the proposed solution (optional)



Onboarding Brief
Set niche, audience, geography, language, platforms, goals and reference styles.



Opportunity Queue
Trends, news, events and ideas ranked into one prioritized list.



3-Option Review
Accept, reject, edit, regenerate or save each sample before it goes live.

Prototype deployed on Google Cloud Run

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Architecture diagram of the proposed solution

Experience	Creator / business owner → ApexGrowth AI app on Google Cloud Run
Signals in	Trends (Search Grounding) · News (industry updates) · Personal Ideas (rough thoughts) · Festivals & Events (holidays, local) · Creative References (format & tone)
AI generation	[redacted]
Human control	Review dashboard · Accept · Reject · Edit · Regenerate · Save
Output	Blotato API → 9 social platforms · Google Sheets → performance metrics

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The Opportunity Pipeline

5 core signals ingested before human review

Trends
Gemini Search Grounding tracks emerging niche conversations and viral hashtag patterns.

News
Scans timely developments and industry updates worth explaining or discussing.

Personal Ideas
Turns rough thoughts, opinions and backlogs into structured posts.

Festivals & Events
Identifies upcoming holidays, cultural occasions and local happenings.

Creative References
Replicates preferred format, tone and structure without plagiarism.

[redacted]

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Technologies to be used in the solution

Google AI

Gemini 1.5 Flash
Rapid multimodal video vision and text

Search Grounding
Tracks trends and hashtag patterns

Google AI Studio
API keys, zero infrastructure cost

Cloud & App

Google Cloud Run
Serverless, high availability, autoscaling

Review dashboard
3-sample human-in-the-loop interface

Google Sheets
Performance metrics log

Publishing

Blotato API
Dispatch to 9 platforms

TikTok · Instagram · YouTube · X · Facebook · LinkedIn · Pinterest · Threads · Bluesky

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Estimated implementation cost (optional)

Architecture stack

Google Cloud Run
Serverless containers: zero server maintenance, high availability and instant autoscaling.

Google AI Studio (Gemini 1.5)
API keys for rapid multimodal video vision and text generation.

Blotato API
Dispatches approved content to 9 social platforms.

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Snapshots of the prototype

Illustrative UI of the 5-stage workflow



Stage 1
Onboarding Brief
Niche, audience, geography, language and goals.



Stage 2
Opportunity Discovery
Trends, news, events and ideas, prioritized.



Stage 3
Multimodal Generation
Gemini builds 3 distinct sample videos.



Stage 4
Human-in-the-Loop
Accept, Reject, Edit, Regenerate or Save.



Stage 5
Distribution & Analytics
9 platforms; metrics logged to Google Sheets.

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Prototype Performance report/Benchmarking

Figures as stated in the project proposal

20+
hours saved weekly for each business owner

5
information signals ingested per cycle

3
distinct sample videos to choose from

9
social platforms reached on approval

Autoscaling on Cloud Run
Zero server maintenance, high availability and instant autoscaling during peak rendering cycles.

Human oversight on every post
Creators Accept, Reject, Edit, Regenerate or Save before anything is dispatched to the platforms.

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Additional Details/Future Development (if any)

Community benefits & value creation

Operational Liberation

Saves owners 20+ hours weekly by automating multi-source data aggregation, trend synthesis and multi-platform scheduling.

Multi-Signal Opportunity Pipeline

Aggregates trends, news, personal ideas, festivals/events and creative references so content is timely, unique and non-derivative.

Supervised Autonomy (HITL)

Pairs AI automation with a 3-sample selection dashboard for brand safety, tone consistency and editorial oversight.

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Provide links to your:

GitHub Public Repository
https://github.com/user/apexgrowth-ai-gcp

Demo Video Link (3 Minutes)
https://youtu.be/sample_apexgrowth_demo

Final Product Link (Google Cloud Run)
https://apexgrowth-ai-xxxx-uc.a.run.app

PDF Pitch Deck Presentation
ApexGrowth_AI_Builders_Cup_Pitch.pdf

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Thank you

